Quiz App with Timer

Purpose:

- To create an interactive quiz application where users answer multiple-choice questions within a time limit per question (or overall).
- To assess or train knowledge in a timed setting.
- As an interview/test assignment, it allows evaluating the candidate's understanding of UI design, state management, timers, user interactions, and responsive layouts.

Scope & Features

Core Features (Must have)

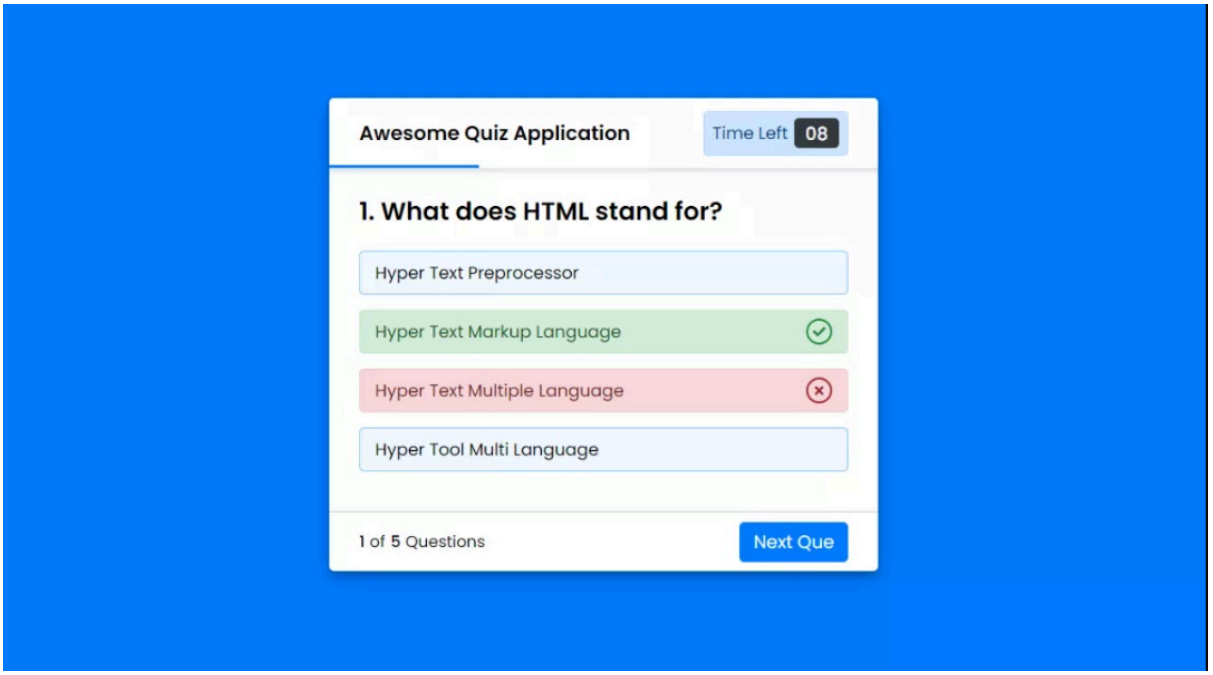
1. **Start / Welcome Screen**
    - A landing view that introduces the quiz and has a "Start Quiz" button.
    - Possibly an info / rules screen before beginning.
2. **Question Display**
    - Show questions one at a time.
    - Each question has 4 options (multiple choice).
    - Show current question number / total questions (e.g., "2 of 6").
3. **Timer**
    - A countdown timer visible to the user (e.g. "Time Left: 08 s").
    - When time runs out, auto-move to next question or treat unanswered as incorrect.
4. **Option Selection & Feedback**
    - The user clicks an option to select.
    - After selection (or on timeout), disable further clicks.
    - Highlight the correct option (green) and incorrect option (red) if the user selected wrong.
5. **Navigation / Flow Control**
    - A "Next" button (or auto-advance after timeout).
    - Optionally, a "Previous" button (if backward navigation is allowed).
    - Flow through all questions until the final one.
6. **Result / Score Screen**
    - After all questions, show the total score (e.g. "You scored X out of Y").
    - Provide a button to restart the quiz or go back to start.
7. **Restart / Retry**
    - Allow restarting from the beginning, resetting all states.

System & UI Design

Screens / Views

1. **Welcome / Start Screen**
    - Title of quiz app

    - “Start Quiz” button
    - (Optional) Info / Rules popup (e.g. “You have 15 seconds per question”)
2. **Quiz Screen**
    - Header: shows question index “Qn of total” and timer (e.g. “Time Left: 10s”)
    - Question text area
    - Options list (buttons)
    - Feedback after selection (color highlight)
    - Navigation controls (Next, Previous, or just Next)
3. **Result Screen**
    - Summary heading (e.g. “Quiz Completed”)
    - Score display: “You scored X / Y”
    - Button: “Restart Quiz” or “Try Again”

UI Components / Elements

- Header / Timer area
- Question container
- Option buttons
- Navigation / control buttons
- Feedback highlight states (correct / incorrect)
- Score / result display
- Hide / show / transition logic for screens